

Brock Allan

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EDUCATION

University of California, Irvine

Irvine, California

Bachelor's of Science in **Computer Science**

Cumulative GPA: **3.6**.

- **Coursework:** Systems & Algorithms, **Digital Logic**, Machine Learning, Multivariable Calculus, Linear Algebra, Data Structure Implementation & Analysis, Computer Vision
- **Distinctions:** CLAP certified, **Chancellor's Excellence** Scholar, Dean's Honor List.

SKILLS & TOOLS

- **Programming:** Python (**PyTorch**, NumPy), C++, Java, C, Assembly (x86), **Prolog**
- **Industry Models:** **Diffusion** Models, SDEdit, **UNet** Architecture, Bilinear Interpolation
- **Leadership Positions:** FFA President, NHS President, MESA Member, **Project Leader**

TECHNICAL EXPERIENCE

Generating New Forms of Informal Expressions

|January 2026 - March 2026|

- Explored the capability of **LLMs** to produce informal or novel expressions, including idiomatic and slang, using **Single-shot**, **Few-shot**, and **Fine-Tuning**.
- **Gemma-3-4b-it**, was fine tuned using **Low Rank Adaptation** and **Cross Entropy Loss**, then evaluated for Uniqueness, Creativity, Parts of speech, and Semantic Fluency.
- Gained hands-on expertise in working with models from **HuggingFace** and tuning those models for use in other scenarios.

Sentiment Analysis of IMDB Movie Reviews

|September 2025 - January 2026|

- **Classified** movie reviews as either positive or negative based solely on textual content.
- Implemented **Bag-of-words** using bigrams, **Unsupervised** learning through K-means ++ Clustering, and a **Random Forest** with **XGBoost** classifiers.
- The model reached **86.9%** testing accuracy after **hyperparameter optimization** and **5-Fold Cross Validation**.

PROFESSIONAL EXPERIENCE

Data Entry Clerk - Shakespeare Motors

|June 2024 - Current|

- Developed **Python-based automated pipelines** to extract statistical insights from visual data, utilizing document analysis tools to generate reports from legal imagery.
- Performed advanced image preprocessing and enhancement using **bilinear interpolation** techniques to optimize visual data for high-fidelity document reproduction.

Computer Consultant & Data Analyst - Freedom Tax Insurance

|May 2020 - June 2024|

- Identified **hardware** issues caused by component failures, modeled using **R**.
- Used **diagnostic** tools to repair and refurbish company devices and **servers**.